

2.5 Assemblies and access modifiers

Assemblies are the fundamental unit of packaging in C# (and the .NET framework in general). An assembly is a collection of C# classes that form a program (if a class contains a Main method) or a library (if no Main method exists). The analog to this in Java is the Jar file which houses collections of .class files.

Access modifiers reflect assemblies as the unit of packaging in C#:

- **public** - visible to everyone
- **protected** - visible to self + all subclasses
- **internal** - visible to self + all classes in the same assembly
- **protected internal** - like internal but also visible to all subclasses (even if they are in different assemblies)
- **private** - visible to self

This is slightly more complicated than Java, but the fundamental rule is the same. Expose as little of your members to as little clients as possible. 